LibJuno Software Requirements Specification (SRS)

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1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for LibJuno, a freestanding C11 embedded systems micro-framework.

This document follows the IEEE 830-1998 standard structure.

1.2. Scope

LibJuno provides data structures, memory management primitives, CRC algorithms, and other utilities for embedded systems development. It requires zero dynamic memory allocation and supports freestanding C11 environments.

1.3. Definitions and Acronyms

Term	Definition
Freestanding	A C implementation that does not require a hosted standard library
DI	Dependency Injection — a pattern where dependencies are provided to a component rather than created internally
Vtable	A struct of function pointers enabling polymorphic dispatch
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix

1.4. References

- IEEE 830-1998: Recommended Practice for Software Requirements Specifications
- LibJuno README and documentation

2. Overall Description

2.1. Product Perspective

LibJuno is a standalone C11 library designed to be used as a dependency in embedded systems projects. It makes no assumptions about the target platform and uses dependency injection to remain adaptable.

2.2. Design Constraints

- No dynamic memory allocation (`malloc`, `calloc`, `realloc`, `free`)
- All memory is caller-owned and injected
- Must compile as freestanding C11 (`-nostdlib -ffreestanding`)
- All warnings treated as errors (`-Werror`)

2.3. Assumptions and Dependencies

- A C11-compliant compiler is available
- CMake ≥ 3.10 is available for building
- Unity test framework is available for testing (build-time only)

3. Specific Requirements

3.1. Module: APP

3.1.1. REQ-APP-001: Application Root Structure

Description

The application root shall embed a module root with an API vtable pointer.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-006, REQ-SYS-007, REQ-SYS-009, REQ-SYS-010

Code Traces

- `include/juno/app/app_api.h:52` (JUNO_MODULE_ROOT)

3.1.2. REQ-APP-002: OnStart Lifecycle Hook

Description

The application API vtable shall define an OnStart operation called once to initialize resources.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-APP-001

Code Traces

- `include/juno/app/app_api.h:58` (JUNO_STATUS_T)

3.1.3. REQ-APP-003: OnProcess Lifecycle Hook

Description

The application API vtable shall define an OnProcess operation that may be invoked repeatedly.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-APP-001

Code Traces

- `include/juno/app/app_api.h:61` (JUNO_STATUS_T)

3.1.4. REQ-APP-004: OnExit Lifecycle Hook

Description

The application API vtable shall define an OnExit operation called once prior to shutdown.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-APP-001

Code Traces

- `include/juno/app/app_api.h:64` (JUNO_STATUS_T)

3.2. Module: ARRAY

3.2.1. REQ-ARRAY-001: Array Initialization

Description

The array module shall initialize from a caller-provided API vtable and capacity.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-006, REQ-SYS-007, REQ-SYS-009

Code Traces

- `include/juno/ds/array_api.h:163` (JunoDs_ArrayInit)

Test Traces

- `tests/test_array.c:150` (test_array_init_success)
- `tests/test_array.c:162` (test_array_init_null_array)

3.2.2. REQ-ARRAY-002: Array SetAt

Description

The array shall copy a value into a slot at a given index in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-ARRAY-001

Code Traces

- `include/juno/ds/array_api.h:82` (JUNO_STATUS_T)

Test Traces

- `tests/test_array.c:174` (test_array_set_at)

3.2.3. REQ-ARRAY-003: Array GetAt

Description

The array shall return a pointer descriptor to the slot at a given index in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-ARRAY-001

Code Traces

- `include/juno/ds/array_api.h:89` (JUNO_RESULT_POINTER_T)

Test Traces

- `tests/test_array.c:184` (test_array_get_at)

3.2.4. REQ-ARRAY-004: Array RemoveAt

Description

The array shall clear the slot at a given index in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-ARRAY-001

Code Traces

- `include/juno/ds/array_api.h:96` (JUNO_STATUS_T)

Test Traces

- `tests/test_array.c:195` (test_array_remove_at)

3.2.5. REQ-ARRAY-005: Array Bounds Checking

Description

The array shall return JUNO_STATUS_OOB_ERROR when an index is outside its capacity.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Code Traces

- `include/juno/ds/array_api.h:140` (JunoDs_ArrayVerifyIndex)

Test Traces

- `tests/test_array.c:208` (test_array_set_at_oob)
- `tests/test_array.c:218` (test_array_get_at_oob)
- `tests/test_array.c:226` (test_array_remove_at_oob)
- `tests/test_queue.c:913` (test_queue_array_api_get_at_out_of_bounds)
- `tests/test_queue.c:923` (test_queue_array_api_remove_at_out_of_bounds)
- `tests/test_stack.c:926` (test_stack_array_api_get_at_out_of_bounds)
- `tests/test_stack.c:936` (test_stack_array_api_remove_at_out_of_bounds)

3.2.6. REQ-ARRAY-006: Array Vtable Validation

Description

The array module shall verify that the array API vtable entries are non-null.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic

allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-010

Code Traces

- `include/juno/ds/array_api.h:120` (JunoDs_ArrayVerify)

Test Traces

- `tests/test_array.c:238` (test_array_verify_null_api)
- `tests/test_queue.c:892` (test_queue_array_api_set_at_invalid_api)
- `tests/test_stack.c:905` (test_stack_array_api_set_at_invalid_api)
- `tests/test_stack.c:946` (test_stack_array_api_null_array)

3.2.7. REQ-ARRAY-007: Array Capacity Validation

Description

The array module shall verify that the array capacity is non-zero.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Code Traces

- `include/juno/ds/array_api.h:120` (JunoDs_ArrayVerify)
- `include/juno/ds/array_api.h:163` (JunoDs_ArrayInit)

Test Traces

- `tests/test_array.c:253` (test_array_verify_zero_capacity)
- `tests/test_array.c:264` (test_array_init_zero_capacity)

3.3. Module: CRC

3.3.1. REQ-CRC-001: CRC Streaming Update

Description

Each CRC function shall accept a running CRC value, data buffer, and size, returning the updated

CRC value.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-009

Code Traces

- `src/juno_arc.c:21` (Juno_CrcArcUpdate)
- `src/juno_binhex.c:21` (Juno_CrcBinhexUpdate)
- `src/juno_ccitt.c:21` (Juno_CrcCcittUpdate)
- `src/juno_ccitt32.c:20` (Juno_CrcCcitt32Update)
- `src/juno_kermit.c:21` (Juno_CrcKermitUpdate)
- `src/juno_zip.c:20` (Juno_CrcZipUpdate)

Test Traces

- `tests/test_crc.c:81` (test_various_crc_behaviour)

3.3.2. REQ-CRC-002: CRC Null Safety

Description

Each CRC function shall return 0 when passed a NULL data pointer.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-005

Code Traces

- `src/juno_arc.c:21` (Juno_CrcArcUpdate)
- `src/juno_binhex.c:21` (Juno_CrcBinhexUpdate)
- `src/juno_ccitt.c:21` (Juno_CrcCcittUpdate)
- `src/juno_ccitt32.c:20` (Juno_CrcCcitt32Update)
- `src/juno_kermit.c:21` (Juno_CrcKermitUpdate)
- `src/juno_zip.c:20` (Juno_CrcZipUpdate)

Test Traces

- `tests/test_crc.c:70` (test_empty_crc_all)

3.3.3. REQ-CRC-003: CRC Zero Length Safety

Description

Each CRC function shall return 0 when passed a zero data size.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-005

Code Traces

- `src/juno_arc.c:21` (Juno_CrcArcUpdate)
- `src/juno_binhex.c:21` (Juno_CrcBinhexUpdate)
- `src/juno_ccitt.c:21` (Juno_CrcCcittUpdate)
- `src/juno_ccitt32.c:20` (Juno_CrcCcitt32Update)
- `src/juno_kermit.c:21` (Juno_CrcKermitUpdate)
- `src/juno_zip.c:20` (Juno_CrcZipUpdate)

Test Traces

- `tests/test_crc.c:70` (test_empty_crc_all)

3.3.4. REQ-CRC-004: CRC-16 ARC

Description

The CRC module shall compute a CRC-16/ARC update with initial seed 0.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-001

Code Traces

- `src/juno_arc.c:21` (Juno_CrcArcUpdate)

Test Traces

- `tests/test_crc.c:81` (test_various_crc_behaviour)

3.3.5. REQ-CRC-005: CRC-16 BinHex

Description

The CRC module shall compute a BinHex 16-bit CRC update with initial seed 0.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-001

Code Traces

- `src/juno_binhex.c:21` (Juno_CrcBinhexUpdate)

Test Traces

- `tests/test_crc.c:81` (test_various_crc_behaviour)

3.3.6. REQ-CRC-006: CRC-16 CCITT

Description

The CRC module shall compute a CRC-16/CCITT update with initial seed 0xFFFF.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-001

Code Traces

- `src/juno_ccitt.c:21` (Juno_CrcCcittUpdate)

Test Traces

- `tests/test_crc.c:56` (test_ccitt_crc)

3.3.7. REQ-CRC-007: CRC-16 Kermit

Description

The CRC module shall compute a CRC-16/Kermit update with initial seed 0.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-001

Code Traces

- `src/juno_kermit.c:21` (Juno_CrcKermitUpdate)

Test Traces

- `tests/test_crc.c:81` (test_various_crc_behaviour)

3.3.8. REQ-CRC-008: CRC-32 CCITT32

Description

The CRC module shall compute a 32-bit CCITT CRC update with initial seed 0xFFFFFFFF.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-001

Code Traces

- `src/juno_ccitt32.c:20` (Juno_CrcCcitt32Update)

Test Traces

- `tests/test_crc.c:81` (test_various_crc_behaviour)

3.3.9. REQ-CRC-009: CRC-32 ZIP

Description

The CRC module shall compute a CRC-32/ZIP update with initial seed 0xFFFFFFFF.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-001

Code Traces

- `src/juno_zip.c:20` (Juno_CrcZipUpdate)

Test Traces

- `tests/test_crc.c:37` (test_zip_crc)

3.3.10. REQ-CRC-010: CRC-32 ZIP Final XOR

Description

The CRC-32/ZIP function shall apply a final XOR of 0xFFFFFFFF to the computed value when data is processed.

Rationale

Enables data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-CRC-009

Code Traces

- `src/juno_zip.c:20` (Juno_CrcZipUpdate)

Test Traces

- `tests/test_crc.c:37` (test_zip_crc)

3.4. Module: HASH

3.4.1. REQ-HASH-001: DJB2 Hash Computation

Description

The hash module shall compute the djb2 hash over a byte buffer with initial seed 5381.

Rationale

Provides data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-008, REQ-SYS-009

Code Traces

- `include/juno/hash/hash_djb2.h:47` (JunoHash_Djb2)

Test Traces

- `tests/test_hash.c:37` (test_hash_djb2_known_value)
- `tests/test_hash.c:52` (test_hash_djb2_empty_buffer)
- `tests/test_hash.c:62` (test_hash_djb2_different_inputs_differ)

3.4.2. REQ-HASH-002: DJB2 Null Buffer Rejection

Description

The hash module shall return JUNO_STATUS_NULLPTR_ERROR when passed a NULL data pointer.

Rationale

Provides data integrity verification for communication and storage in embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Code Traces

- `include/juno/hash/hash_djb2.h:47` (JunoHash_Djb2)

Test Traces

- `tests/test_hash.c:78` (test_hash_djb2_null_buffer)

3.4.3. REQ-HASH-003: DJB2 Result Type

Description

The hash module shall return the hash value in a result type combining a status code with a size_t payload.

Rationale

Provides data integrity verification for communication and storage in embedded systems.

Verification Method

Inspection

Uses (parent)

REQ-SYS-004

Code Traces

- `include/juno/hash/hash_djb2.h:47` (JunoHash_Djb2)

3.5. Module: HEAP

3.5.1. REQ-HEAP-001: Heap Initialization

Description

The heap module shall initialize from a caller-provided backing array.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic

allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-007, REQ-SYS-008, REQ-SYS-009, REQ-ARRAY-001

Code Traces

- `src/juno_heap.c:32` (JunoDs_Heap_Init)

Test Traces

- `tests/test_heap.c:353` (test_heap_init_nominal)
- `tests/test_heap.c:363` (test_heap_init_null_heap)
- `tests/test_heap.c:373` (test_heap_init_null_api)
- `tests/test_heap.c:383` (test_heap_init_zero_capacity)

3.5.2. REQ-HEAP-002: Heap Element Comparison Injection

Description

The heap module shall accept an injected element pointer API with Compare and Swap operations.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-007, REQ-HEAP-001

Code Traces

- `src/juno_heap.c:32` (JunoDs_Heap_Init)

Test Traces

- `tests/test_heap.c:526` (test_heap_min_heap_insert_and_property)
- `tests/test_heap.c:709` (test_heap_update_with_compare_error)
- `tests/test_heap.c:727` (test_heap_update_with_swap_error)
- `tests/test_heap.c:825` (test_heap_siftdown_with_compare_error)
- `tests/test_heap.c:844` (test_heap_siftdown_with_swap_error)
- `tests/test_heap.c:909` (test_heap_compare_function_error)
- `tests/test_heap.c:931` (test_heap_swap_function_error)

- `tests/test_heap.c:994` (test_heap_mixed_min_max_operations)

3.5.3. REQ-HEAP-003: Heap Insert

Description

The heap shall insert an element and restore the heap property in O(log n) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-HEAP-001

Code Traces

- `src/juno_heap.c:258` (JunoDs_Heap_Insert)

Test Traces

- `tests/test_heap.c:415` (test_heap_insert_single_item_max_heap)
- `tests/test_heap.c:432` (test_heap_insert_multiple_items_max_heap)
- `tests/test_heap.c:453` (test_heap_insert_to_full_capacity)
- `tests/test_heap.c:472` (test_heap_insert_overflow)
- `tests/test_heap.c:495` (test_heap_insert_null_heap)
- `tests/test_heap.c:505` (test_heap_insert_invalid_pointer)
- `tests/test_heap.c:526` (test_heap_min_heap_insert_and_property)
- `tests/test_heap.c:653` (test_heap_update_single_element)
- `tests/test_heap.c:669` (test_heap_update_bubble_up)
- `tests/test_heap.c:691` (test_heap_update_empty_heap)
- `tests/test_heap.c:702` (test_heap_update_null_heap)
- `tests/test_heap.c:957` (test_heap_stress_insert_pop_cycles)
- `tests/test_heap.c:994` (test_heap_mixed_min_max_operations)
- `tests/test_heap.c:1033` (test_heap_single_element_capacity)
- `tests/test_heap.c:1058` (test_heap_data_integrity_after_operations)

3.5.4. REQ-HEAP-004: Heap Pop

Description

The heap shall remove the root element and restore the heap property in O(log n) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-HEAP-001

Code Traces

- `src/juno_heap.c:325` (JunoDs_Heap_Pop)

Test Traces

- `tests/test_heap.c:550` (test_heap_pop_single_item_max_heap)
- `tests/test_heap.c:574` (test_heap_pop_multiple_items_max_heap)
- `tests/test_heap.c:610` (test_heap_pop_empty_heap)
- `tests/test_heap.c:622` (test_heap_pop_null_heap)
- `tests/test_heap.c:631` (test_heap_pop_invalid_pointer)
- `tests/test_heap.c:749` (test_heap_siftdown_from_root)
- `tests/test_heap.c:767` (test_heap_siftdown_from_middle)
- `tests/test_heap.c:787` (test_heap_siftdown_leaf_node)
- `tests/test_heap.c:807` (test_heap_siftdown_empty_heap)
- `tests/test_heap.c:818` (test_heap_siftdown_null_heap)
- `tests/test_heap.c:957` (test_heap_stress_insert_pop_cycles)
- `tests/test_heap.c:994` (test_heap_mixed_min_max_operations)
- `tests/test_heap.c:1033` (test_heap_single_element_capacity)
- `tests/test_heap.c:1058` (test_heap_data_integrity_after_operations)

3.5.5. REQ-HEAP-005: Heap Pop Output

Description

The heap Pop operation shall copy the removed root element into a caller-provided pointer descriptor.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-HEAP-004

Code Traces

- `src/juno_heap.c:325` (JunoDs_Heap_Pop)

Test Traces

- `tests/test_heap.c:550` (test_heap_pop_single_item_max_heap)
- `tests/test_heap.c:574` (test_heap_pop_multiple_items_max_heap)
- `tests/test_heap.c:1058` (test_heap_data_integrity_after_operations)

3.5.6. REQ-HEAP-006: Heap Heapify

Description

The heap shall transform the underlying array into a valid heap in O(n) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-HEAP-001

Code Traces

- `src/juno_heap.c:284` (JunoDs_Heap_Heapify)

Test Traces

- `tests/test_heap.c:867` (test_heap_heapify_from_unsorted_array)
- `tests/test_heap.c:886` (test_heap_heapify_empty_heap)
- `tests/test_heap.c:898` (test_heap_heapify_null_heap)

3.5.7. REQ-HEAP-007: Heap Vtable Validation

Description

The heap module shall verify that the heap API vtable entries are non-null before each operation.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-010

Code Traces

- `include/juno/ds/heap_api.h:227` (JunoDs_Heap_Verify)

Test Traces

- `tests/test_heap.c:397` (test_heap_verify_null_heap)
- `tests/test_heap.c:403` (test_heap_verify_invalid_api)

3.5.8. REQ-HEAP-008: Heap Dependency Validation

Description

The heap module shall verify that the backing array and heap pointer API dependencies are valid before each operation.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-011

Code Traces

- `include/juno/ds/heap_api.h:227` (JunoDs_Heap_Verify)

Test Traces

- `tests/test_heap.c:373` (test_heap_init_null_api)
- `tests/test_heap.c:383` (test_heap_init_zero_capacity)

3.6. Module: IO

3.6.1. REQ-IO-001: Async IO Root Structure

Description

The async IO root shall embed a module root with an API vtable pointer.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-006, REQ-SYS-007, REQ-SYS-009, REQ-SYS-010

Code Traces

- `include/juno/io/async_io_api.h:47` (JUNO_MODULE_ROOT)

3.6.2. REQ-IO-002: Async IO Read

Description

The async IO API vtable shall define a Read operation that reads up to the requested number of bytes and updates the size with bytes actually read.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:53` (JUNO_STATUS_T)

3.6.3. REQ-IO-003: Async IO TryRead

Description

The async IO API vtable shall define a TryRead operation that reads with a timeout in microseconds.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:56` (JUNO_STATUS_T)

3.6.4. REQ-IO-004: Async IO ReadUntil

Description

The async IO API vtable shall define a ReadUntil operation that reads until any stop character is received.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:59` (JUNO_STATUS_T)

3.6.5. REQ-IO-005: Async IO TryReadUntil

Description

The async IO API vtable shall define a TryReadUntil operation that reads until a stop character or timeout.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:62` (JUNO_STATUS_T)

3.6.6. REQ-IO-006: Async IO Write

Description

The async IO API vtable shall define a Write operation that writes up to the requested number of bytes and updates the size with bytes actually written.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:65` (JUNO_STATUS_T)

3.6.7. REQ-IO-007: Async IO TryWrite

Description

The async IO API vtable shall define a TryWrite operation that writes with a timeout in microseconds.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:68` (JUNO_STATUS_T)

3.6.8. REQ-IO-008: Async IO Poll

Description

The async IO API vtable shall define a Poll operation that checks for input readiness within a timeout.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-001

Code Traces

- `include/juno/io/async_io_api.h:71` (JUNO_STATUS_T)

3.6.9. REQ-IO-009: I2C IO Root Structure

Description

The I2C IO root shall embed a module root with an API vtable pointer.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-SYS-006

Code Traces

- `include/juno/io/i2c_io_api.h:108` (JUNO_MODULE_ROOT)

3.6.10. REQ-IO-010: I2C Transfer Operation

Description

The I2C IO API vtable shall define a Transfer operation that processes an array of read/write messages.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-009

Code Traces

- `include/juno/io/i2c_io_api.h:134` (JUNO_STATUS_T)

3.6.11. REQ-IO-011: I2C Message Type Discrimination

Description

Each I2C message shall contain a header with a type field distinguishing read from write messages.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-010

Code Traces

- `include/juno/io/i2c_io_api.h:58` (<unknown>)

3.6.12. REQ-IO-012: I2C Device Address

Description

Each I2C message header shall contain the target device address.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-011

Code Traces

- `include/juno/io/i2c_io_api.h:58` (<unknown>)

3.6.13. REQ-IO-013: SPI IO Root Structure

Description

The SPI IO root shall embed a module root with an API vtable pointer.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-SYS-006

Code Traces

- `include/juno/io/spi_io_api.h:46` (JUNO_MODULE_ROOT)

3.6.14. REQ-IO-014: SPI Transaction Operation

Description

The SPI IO API vtable shall define a Transaction operation for full-duplex read/write.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-IO-013

Code Traces

- `include/juno/io/spi_io_api.h:58` (JUNO_STATUS_T)

3.7. Module: LOG

3.7.1. REQ-LOG-001: Log Root Structure

Description

The log root shall embed a module root with an API vtable pointer.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-006, REQ-SYS-009, REQ-SYS-010

Code Traces

- `include/juno/log/log_api.h:51` (JUNO_MODULE_ROOT)

3.7.2. REQ-LOG-002: Debug Level Logging

Description

The log API vtable shall define a LogDebug operation for debug-level messages.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-LOG-001

Code Traces

- `include/juno/log/log_api.h:57` (JUNO_STATUS_T)

3.7.3. REQ-LOG-003: Info Level Logging

Description

The log API vtable shall define a LogInfo operation for info-level messages.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-LOG-001

Code Traces

- `include/juno/log/log_api.h:60` (JUNO_STATUS_T)

3.7.4. REQ-LOG-004: Warning Level Logging

Description

The log API vtable shall define a LogWarning operation for warning-level messages.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-LOG-001

Code Traces

- `include/juno/log/log_api.h:63` (JUNO_STATUS_T)

3.7.5. REQ-LOG-005: Error Level Logging

Description

The log API vtable shall define a LogError operation for error-level messages.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-LOG-001

Code Traces

- `include/juno/log/log_api.h:66` (JUNO_STATUS_T)

3.7.6. REQ-LOG-006: Log Initialization

Description

The log module shall initialize with an injected API vtable and failure handler.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Test

Uses (parent)

REQ-SYS-002, REQ-SYS-007, REQ-LOG-001

Code Traces

- `include/juno/log/log_api.h:71` (JunoLog_LogInit)

Test Traces

- `tests/test_log.c:94` (test_log_init_success)
- `tests/test_log.c:102` (test_log_init_null_root)

3.7.7. REQ-LOG-007: Variadic Message Format

Description

Each log operation shall accept a printf-like format string with variadic arguments.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-LOG-001

Code Traces

- `include/juno/log/log_api.h:57` (JUNO_STATUS_T)
- `include/juno/log/log_api.h:60` (JUNO_STATUS_T)
- `include/juno/log/log_api.h:63` (JUNO_STATUS_T)
- `include/juno/log/log_api.h:66` (JUNO_STATUS_T)

3.8. Module: MAP

3.8.1. REQ-MAP-001: Map Initialization

Description

The map module shall initialize from a caller-provided backing array.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-007, REQ-SYS-008, REQ-SYS-009, REQ-ARRAY-001

Code Traces

- `src/juno_map.c:34` (JunoDs_MapInit)

Test Traces

- `tests/test_map.c:322` (test_map_init_nominal)
- `tests/test_map.c:336` (test_map_init_null_map)
- `tests/test_map.c:343` (test_map_init_null_api)
- `tests/test_map.c:351` (test_map_init_zero_capacity)

3.8.2. REQ-MAP-002: Map Hash Function Injection

Description

The map module shall accept an injected hashable pointer API with Hash and IsValueNull operations.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-007, REQ-MAP-001

Code Traces

- `src/juno_map.c:34` (JunoDs_MapInit)

Test Traces

- `tests/test_map.c:322` (test_map_init_nominal)
- `tests/test_map.c:901` (test_map_entry_hash_nominal)
- `tests/test_map.c:912` (test_map_entry_is_null_true)
- `tests/test_map.c:923` (test_map_entry_is_null_false)

3.8.3. REQ-MAP-003: Map Equality Injection

Description

The map module shall accept an injected value pointer API with an Equals operation for key comparison.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic

allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-007, REQ-MAP-001

Code Traces

- `src/juno_map.c:34` (JunoDs_MapInit)

Test Traces

- `tests/test_map.c:322` (test_map_init_nominal)
- `tests/test_map.c:873` (test_map_entry_equals_same_key)
- `tests/test_map.c:887` (test_map_entry_equals_different_key)

3.8.4. REQ-MAP-004: Map Set

Description

The map shall insert or update a key-value pair using linear probing.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-MAP-001

Code Traces

- `src/juno_map.c:135` (JunoDs_MapSet)

Test Traces

- `tests/test_map.c:365` (test_map_set_single_item)
- `tests/test_map.c:384` (test_map_set_multiple_items)
- `tests/test_map.c:406` (test_map_set_update_existing)
- `tests/test_map.c:430` (test_map_set_with_collision)
- `tests/test_map.c:455` (test_map_set_null_map)
- `tests/test_map.c:663` (test_map_multiple_collisions_linear_probing)
- `tests/test_map.c:714` (test_map_wraparound_probing)
- `tests/test_map.c:741` (test_map_single_element_capacity)
- `tests/test_map.c:776` (test_map_set_get_remove_cycle)

3.8.5. REQ-MAP-005: Map Table Full Rejection

Description

The map shall return JUNO_STATUS_TABLE_FULL_ERROR when no empty slot exists during a Set operation.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-MAP-004

Code Traces

- `src/juno_map.c:135` (JunoDs_MapSet)

Test Traces

- `tests/test_map.c:465` (test_map_set_table_full)

3.8.6. REQ-MAP-006: Map Get

Description

The map shall retrieve an entry by key using linear probing.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-MAP-001

Code Traces

- `src/juno_map.c:109` (JunoDs_MapGet)

Test Traces

- `tests/test_map.c:490` (test_map_get_existing_item)
- `tests/test_map.c:528` (test_map_get_after_collision)
- `tests/test_map.c:556` (test_map_get_null_map)
- `tests/test_map.c:663` (test_map_multiple_collisions_linear_probing)
- `tests/test_map.c:714` (test_map_wraparound_probing)
- `tests/test_map.c:741` (test_map_single_element_capacity)

- `tests/test_map.c:776` (test_map_set_get_remove_cycle)

3.8.7. REQ-MAP-007: Map Key Not Found

Description

The map shall return JUNO_STATUS_DNE_ERROR when a Get operation does not find the key.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-MAP-006

Code Traces

- `src/juno_map.c:109` (JunoDs_MapGet)

Test Traces

- `tests/test_map.c:515` (test_map_get_nonexistent_item)

3.8.8. REQ-MAP-008: Map Remove

Description

The map shall remove an entry by key using linear probing, resetting the slot.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-MAP-001

Code Traces

- `src/juno_map.c:156` (JunoDs_MapRemove)

Test Traces

- `tests/test_map.c:570` (test_map_remove_existing_item)
- `tests/test_map.c:593` (test_map_remove_nonexistent_item)
- `tests/test_map.c:620` (test_map_remove_after_collision)
- `tests/test_map.c:649` (test_map_remove_null_map)
- `tests/test_map.c:776` (test_map_set_get_remove_cycle)

3.8.9. REQ-MAP-009: Map Vtable Validation

Description

The map module shall verify that its API vtable entries are non-null before each operation.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-010

Code Traces

- `include/juno/ds/map_api.h:130` (JunoDs_MapVerify)

Test Traces

- `tests/test_map.c:810` (test_map_verify_valid_map)
- `tests/test_map.c:820` (test_map_verify_null_map)

3.8.10. REQ-MAP-010: Map Dependency Validation

Description

The map module shall verify that the backing array, hashable pointer API, and value pointer API dependencies are valid before each operation.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-011

Code Traces

- `include/juno/ds/map_api.h:130` (JunoDs_MapVerify)

Test Traces

- `tests/test_map.c:343` (test_map_init_null_api)
- `tests/test_map.c:351` (test_map_init_zero_capacity)

3.9. Module: MATH

3.9.1. REQ-MATH-001: Math Constants

Description

The math module shall define double-precision constants for π, π/2, π/4, and e.

Rationale

Provides core math primitives for embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-009

Code Traces

- `include/juno/math/juno_math_constants.h:33` (Pi)

3.9.2. REQ-MATH-002: Vector Types

Description

The math module shall define 2D and 3D vector types in double, float, and int32 precisions.

Rationale

Provides core math primitives for embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-003

Code Traces

- `include/juno/math/juno_vec_types.h:64` (<unknown>)

3.9.3. REQ-MATH-003: Vector Coordinate Access

Description

Each vector type shall support both named Cartesian field access and raw array indexing.

Rationale

Provides core math primitives for embedded software.

Verification Method

Inspection

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec_types.h:64` (<unknown>)

3.9.4. REQ-MATH-004: Quaternion Type

Description

The math module shall define a real quaternion type with [s, i, j, k] component order in double precision.

Rationale

Provides core math primitives for embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-003

Code Traces

- `include/juno/math/juno_vec_types.h:674` (<unknown>)

3.9.5. REQ-MATH-005: Vector Addition

Description

The math module shall compute component-wise vector addition for each vector type.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:67` (Juno_Vec2_F64_Add)
- `include/juno/math/juno_vec.h:139` (Juno_Vec2_F32_Add)
- `include/juno/math/juno_vec.h:201` (Juno_Vec2_I32_Add)
- `include/juno/math/juno_vec.h:264` (Juno_Vec3_F64_Add)
- `include/juno/math/juno_vec.h:335` (Juno_Vec3_F32_Add)
- `include/juno/math/juno_vec.h:405` (Juno_Vec3_I32_Add)

Test Traces

- `tests/test_math.c:35` (test_vec2_f64_add)
- `tests/test_math.c:102` (test_vec2_f32_add)

- `tests/test_math.c:169` (test_vec2_i32_add)
- `tests/test_math.c:236` (test_vec3_f64_add)
- `tests/test_math.c:306` (test_vec3_f32_add)
- `tests/test_math.c:376` (test_vec3_i32_add)

3.9.6. REQ-MATH-006: Vector Subtraction

Description

The math module shall compute component-wise vector subtraction for each vector type.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:82` (Juno_Vec2_F64_Sub)
- `include/juno/math/juno_vec.h:151` (Juno_Vec2_F32_Sub)
- `include/juno/math/juno_vec.h:213` (Juno_Vec2_I32_Sub)
- `include/juno/math/juno_vec.h:277` (Juno_Vec3_F64_Sub)
- `include/juno/math/juno_vec.h:348` (Juno_Vec3_F32_Sub)
- `include/juno/math/juno_vec.h:418` (Juno_Vec3_I32_Sub)

Test Traces

- `tests/test_math.c:48` (test_vec2_f64_sub)
- `tests/test_math.c:115` (test_vec2_f32_sub)
- `tests/test_math.c:182` (test_vec2_i32_sub)
- `tests/test_math.c:249` (test_vec3_f64_sub)
- `tests/test_math.c:319` (test_vec3_f32_sub)
- `tests/test_math.c:389` (test_vec3_i32_sub)

3.9.7. REQ-MATH-007: Vector Scalar Multiplication

Description

The math module shall compute component-wise scalar multiplication for each vector type.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:96` (Juno_Vec2_F64_Mult)
- `include/juno/math/juno_vec.h:162` (Juno_Vec2_F32_Mult)
- `include/juno/math/juno_vec.h:224` (Juno_Vec2_I32_Mult)
- `include/juno/math/juno_vec.h:289` (Juno_Vec3_F64_Mult)
- `include/juno/math/juno_vec.h:360` (Juno_Vec3_F32_Mult)
- `include/juno/math/juno_vec.h:430` (Juno_Vec3_I32_Mult)

Test Traces

- `tests/test_math.c:61` (test_vec2_f64_mult)
- `tests/test_math.c:128` (test_vec2_f32_mult)
- `tests/test_math.c:195` (test_vec2_i32_mult)
- `tests/test_math.c:262` (test_vec3_f64_mult)
- `tests/test_math.c:332` (test_vec3_f32_mult)
- `tests/test_math.c:402` (test_vec3_i32_mult)

3.9.8. REQ-MATH-008: Vector Dot Product

Description

The math module shall compute the dot product for each vector type.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:108` (Juno_Vec2_F64_Dot)
- `include/juno/math/juno_vec.h:173` (Juno_Vec2_F32_Dot)
- `include/juno/math/juno_vec.h:235` (Juno_Vec2_I32_Dot)
- `include/juno/math/juno_vec.h:301` (Juno_Vec3_F64_Dot)
- `include/juno/math/juno_vec.h:372` (Juno_Vec3_F32_Dot)

- `include/juno/math/juno_vec.h:442` (Juno_Vec3_I32_Dot)

Test Traces

- `tests/test_math.c:73` (test_vec2_f64_dot)
- `tests/test_math.c:140` (test_vec2_f32_dot)
- `tests/test_math.c:207` (test_vec2_i32_dot)
- `tests/test_math.c:274` (test_vec3_f64_dot)
- `tests/test_math.c:344` (test_vec3_f32_dot)
- `tests/test_math.c:414` (test_vec3_i32_dot)

3.9.9. REQ-MATH-009: 2D Cross Product

Description

The math module shall compute the pseudoscalar z-component of the 2D cross product.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:119` (Juno_Vec2_F64_Cross)
- `include/juno/math/juno_vec.h:182` (Juno_Vec2_F32_Cross)
- `include/juno/math/juno_vec.h:244` (Juno_Vec2_I32_Cross)

Test Traces

- `tests/test_math.c:83` (test_vec2_f64_cross)
- `tests/test_math.c:150` (test_vec2_f32_cross)
- `tests/test_math.c:217` (test_vec2_i32_cross)

3.9.10. REQ-MATH-010: 3D Cross Product

Description

The math module shall compute the 3D cross product returning an orthogonal vector.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:311` (Juno_Vec3_F64_Cross)
- `include/juno/math/juno_vec.h:381` (Juno_Vec3_F32_Cross)
- `include/juno/math/juno_vec.h:451` (Juno_Vec3_I32_Cross)

Test Traces

- `tests/test_math.c:284` (test_vec3_f64_cross)
- `tests/test_math.c:354` (test_vec3_f32_cross)
- `tests/test_math.c:424` (test_vec3_i32_cross)

3.9.11. REQ-MATH-011: Squared L2 Norm

Description

The math module shall compute the squared L2 norm for each vector type.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-002

Code Traces

- `include/juno/math/juno_vec.h:129` (Juno_Vec2_F64_L2Norm2)
- `include/juno/math/juno_vec.h:191` (Juno_Vec2_F32_L2Norm2)
- `include/juno/math/juno_vec.h:254` (Juno_Vec2_I32_L2Norm2)
- `include/juno/math/juno_vec.h:325` (Juno_Vec3_F64_L2Norm2)
- `include/juno/math/juno_vec.h:395` (Juno_Vec3_F32_L2Norm2)
- `include/juno/math/juno_vec.h:465` (Juno_Vec3_I32_L2Norm2)

Test Traces

- `tests/test_math.c:93` (test_vec2_f64_L2)
- `tests/test_math.c:160` (test_vec2_f32_L2)
- `tests/test_math.c:227` (test_vec2_i32_L2)
- `tests/test_math.c:297` (test_vec3_f64_L2)
- `tests/test_math.c:367` (test_vec3_f32_L2)
- `tests/test_math.c:437` (test_vec3_i32_L2)

3.9.12. REQ-MATH-012: Quaternion Addition

Description

The math module shall compute component-wise quaternion addition.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-004

Code Traces

- `include/juno/math/juno_vec.h:476` (Juno_RQuat_F64_Add)

Test Traces

- `tests/test_math.c:446` (test_rquat_f64_add)

3.9.13. REQ-MATH-013: Quaternion Subtraction

Description

The math module shall compute component-wise quaternion subtraction.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-004

Code Traces

- `include/juno/math/juno_vec.h:490` (Juno_RQuat_F64_Sub)

Test Traces

- `tests/test_math.c:458` (test_rquat_f64_sub)

3.9.14. REQ-MATH-014: Quaternion Scalar Multiplication

Description

The math module shall compute component-wise quaternion scalar multiplication.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-004

Code Traces

- `include/juno/math/juno_vec.h:503` (Juno_RQuat_F64_Mult)

Test Traces

- `tests/test_math.c:470` (test_rquat_f64_mult_scalar)

3.9.15. REQ-MATH-015: Hamilton Product

Description

The math module shall compute the right-handed Hamilton product of two quaternions.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-004

Code Traces

- `include/juno/math/juno_vec.h:523` (Juno_RQuat_F64_HamProd)

Test Traces

- `tests/test_math.c:482` (test_rquat_f64_hamprod)
- `tests/test_math.c:495` (test_rquat_f64_hamprod_general)

3.9.16. REQ-MATH-016: Quaternion Conjugate

Description

The math module shall compute the conjugate of a quaternion by negating the vector part.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-004

Code Traces

- `include/juno/math/juno_vec.h:539` (Juno_RQuat_F64_Conj)

Test Traces

- `tests/test_math.c:509` (test_rquat_f64_conj)

3.9.17. REQ-MATH-017: Quaternion Reciprocal

Description

The math module shall compute the multiplicative inverse of a quaternion as conj(q) divided by the squared norm.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-016, REQ-MATH-018

Code Traces

- `include/juno/math/juno_vec.h:562` (Juno_RQuat_F64_Recip)

Test Traces

- `tests/test_math.c:538` (test_rquat_f64_recip)

3.9.18. REQ-MATH-018: Quaternion Squared Norm

Description

The math module shall compute the squared L2 norm of a quaternion.

Rationale

Provides core math primitives for embedded software.

Verification Method

Test

Uses (parent)

REQ-MATH-004

Code Traces

- `include/juno/math/juno_vec.h:551` (Juno_RQuat_F64_L2Norm2)

Test Traces

- `tests/test_math.c:520` (test_rquat_f64_l2norm2)
- `tests/test_math.c:529` (test_rquat_f64_l2norm)

3.10. Module: MEMORY

3.10.1. REQ-MEMORY-001: Memory Allocator Interface

Description

The memory module shall define an abstract allocator API with Get, Update, and Put operations.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-009, REQ-SYS-010

Code Traces

- `include/juno/memory/memory_api.h:80` (<unknown>)

3.10.2. REQ-MEMORY-002: Allocator Pointer API Dependency

Description

The memory allocator root shall hold an injected pointer API for internal memory operations.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Inspection

Uses (parent)

REQ-SYS-007, REQ-POINTER-001

Code Traces

- `include/juno/memory/memory_api.h:72` (JUNO_MODULE_ROOT)

3.10.3. REQ-MEMORY-003: Block Allocator Initialization

Description

The block allocator shall initialize from caller-provided backing memory, metadata array, type size, alignment, and length.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-002, REQ-SYS-007, REQ-SYS-008, REQ-SYS-011, REQ-MEMORY-001

Code Traces

- `src/juno_memory_block.c:248` (JunoMemory_BlockInit)

Test Traces

- `tests/test_memory.c:272` (test_invalid_init_parameters)
- `tests/test_memory.c:428` (test_bad_api)

3.10.4. REQ-MEMORY-004: Block Allocator Get

Description

The block allocator shall allocate a block of the configured element size in O(1) time.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-001

Code Traces

- `src/juno_memory_block.c:65` (Juno_MemoryBlkGet)

Test Traces

- `tests/test_memory.c:83` (test_nominal_single_alloc_and_free)
- `tests/test_memory.c:123` (test_nominal_multiple_alloc_and_free)
- `tests/test_memory.c:185` (test_negative_memory_empty)
- `tests/test_memory.c:234` (test_negative_memory_full)
- `tests/test_memory.c:373` (test_generic_memory_get_put)
- `tests/test_memory.c:406` (test_zero_size_allocation)

3.10.5. REQ-MEMORY-005: Block Allocator Free Block Reuse

Description

The block allocator shall reuse freed blocks via a LIFO free list when available.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded

systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-004

Code Traces

- `src/juno_memory_block.c:65` (Juno_MemoryBlkGet)

Test Traces

- `tests/test_memory.c:123` (test_nominal_multiple_alloc_and_free)

3.10.6. REQ-MEMORY-006: Block Allocator Put

Description

The block allocator shall free a previously allocated block in O(1) time.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-001

Code Traces

- `src/juno_memory_block.c:158` (Juno_MemoryBlkPut)

Test Traces

- `tests/test_memory.c:83` (test_nominal_single_alloc_and_free)
- `tests/test_memory.c:123` (test_nominal_multiple_alloc_and_free)
- `tests/test_memory.c:373` (test_generic_memory_get_put)

3.10.7. REQ-MEMORY-007: Block Allocator Double Free Rejection

Description

The block allocator shall reject double frees with an error status.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-006

Code Traces

- `src/juno_memory_block.c:158` (Juno_MemoryBlkPut)

Test Traces

- `tests/test_memory.c:290` (test_double_free)

3.10.8. REQ-MEMORY-008: Block Allocator Invalid Address Rejection

Description

The block allocator shall reject frees of invalid or misaligned addresses with an error status.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-006

Code Traces

- `src/juno_memory_block.c:158` (Juno_MemoryBlkPut)

Test Traces

- `tests/test_memory.c:319` (test_free_unallocated)
- `tests/test_memory.c:464` (test_invalid_size_and_addr)

3.10.9. REQ-MEMORY-009: Block Allocator Update

Description

The block allocator shall adjust a descriptor's size in-place up to the configured element size.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-001

Code Traces

- `src/juno_memory_block.c:139` (Juno_MemoryBlkUpdate)

Test Traces

- `tests/test_memory.c:341` (test_update_memory)

3.10.10. REQ-MEMORY-010: Block Allocator No Memory Move

Description

The block allocator Update shall not move memory when adjusting descriptor size.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-MEMORY-009

Code Traces

- `src/juno_memory_block.c:139` (Juno_MemoryBlkUpdate)

Test Traces

- `tests/test_memory.c:341` (test_update_memory)

3.11. Module: MODULE

3.11.1. REQ-MODULE-001: Module Root Structure

Description

The module system shall define a module root struct containing an API vtable pointer.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-006, REQ-SYS-009

Code Traces

- `include/juno/module.h:128` (<unknown>)

3.11.2. REQ-MODULE-002: Module Root Failure Handler

Description

The module root shall contain a failure handler pointer and user data pointer.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-STATUS-003

Code Traces

- `include/juno/module.h:128` (<unknown>)

3.11.3. REQ-MODULE-003: Module Derivation

Description

Derived module structs shall embed the root as their first member, enabling safe upcasting to the root type.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-006

Code Traces

- `include/juno/module.h:160` (<unknown>)

3.11.4. REQ-MODULE-004: Module Union

Description

A module union shall contain the root and all derivation types for polymorphic storage.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-006

Code Traces

- `include/juno/module.h:111` (<unknown>)

3.11.5. REQ-MODULE-005: Result Type

Description

The module system shall define a result type combining a status code with a typed success payload.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-004

Code Traces

- `include/juno/module.h:197` (<unknown>)

3.11.6. REQ-MODULE-006: Option Type

Description

The module system shall define an option type combining a presence flag with a typed payload.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-004

Code Traces

- `include/juno/module.h:241` (<unknown>)

3.12. Module: MTX

3.12.1. REQ-MTX-001: Mutex Root Structure

Description

The mutex root shall embed a module root with an API vtable pointer.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-006, REQ-SYS-007, REQ-SYS-009

Code Traces

- `include/juno/mp/mtx_api.h:44` (JUNO_MODULE_ROOT)

3.12.2. REQ-MTX-002: TryLock Operation

Description

The mutex API vtable shall define a TryLock operation that attempts to acquire the lock and returns immediately with a boolean result.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-MTX-001

Code Traces

- `include/juno/mp/mtx_api.h:50` (JUNO_RESULT_BOOL_T)

3.12.3. REQ-MTX-003: Lock Operation

Description

The mutex API vtable shall define a Lock operation that blocks until the lock is acquired.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-MTX-001

Code Traces

- `include/juno/mp/mtx_api.h:53` (JUNO_STATUS_T)

3.12.4. REQ-MTX-004: Free Operation

Description

The mutex API vtable shall define a Free operation that releases the lock.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Inspection

Uses (parent)

REQ-MTX-001

Code Traces

- `include/juno/mp/mtx_api.h:56` (JUNO_STATUS_T)

3.12.5. REQ-MTX-005: Mutex Vtable Validation

Description

The mutex module shall verify that its API vtable and all function pointers within it are non-NULL.

Rationale

Provides a portable abstraction interface for platform-specific capabilities.

Verification Method

Test

Uses (parent)

REQ-SYS-010

Code Traces

- `include/juno/mp/mtx_api.h:61` (JunoMp_MtxVerify)

Test Traces

- `tests/test_mtx.c:77` (test_mtx_verify_valid)
- `tests/test_mtx.c:84` (test_mtx_verify_null_root)
- `tests/test_mtx.c:91` (test_mtx_verify_null_api)
- `tests/test_mtx.c:99` (test_mtx_verify_null_trylock)

3.13. Module: POINTER

3.13.1. REQ-POINTER-001: Pointer Descriptor Structure

Description

The pointer module shall define a pointer trait (JUNO_POINTER_T) containing an API vtable pointer, memory address, size, and alignment.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-009

Code Traces

- `include/juno/memory/pointer_api.h:55` (JUNO_TRAIT_ROOT)

3.13.2. REQ-POINTER-002: Pointer Copy

Description

The pointer API shall copy the value referenced by a source descriptor into a destination descriptor's storage location.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-POINTER-001

Code Traces

- `include/juno/memory/pointer_api.h:156` (JunoMemory_PointerApiVerify)

Test Traces

- `tests/test_map.c:833` (test_map_entry_copy_nominal)
- `tests/test_map.c:849` (test_map_entry_copy_null_dest)
- `tests/test_pointer.c:89` (test_pointer_copy_success)
- `tests/test_pointer.c:100` (test_pointer_copy_null_src_addr)
- `tests/test_stack.c:1032` (test_stack_pointer_copy_invalid_dest)

- `tests/test_stack.c:1043` (test_stack_pointer_copy_invalid_src)

3.13.3. REQ-POINTER-003: Pointer Reset

Description

The pointer API shall reset the memory at a pointer descriptor to a defined empty state.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-POINTER-001

Code Traces

- `include/juno/memory/pointer_api.h:156` (JunoMemory_PointerApiVerify)

Test Traces

- `tests/test_map.c:860` (test_map_entry_reset_nominal)
- `tests/test_pointer.c:113` (test_pointer_reset_success)
- `tests/test_pointer.c:123` (test_pointer_reset_null_addr)
- `tests/test_stack.c:1054` (test_stack_pointer_reset_invalid)
- `tests/test_stack.c:1062` (test_stack_pointer_reset_valid)

3.13.4. REQ-POINTER-004: Pointer Descriptor Validation

Description

The pointer module shall validate that a pointer descriptor's API, address, and size are non-null/non-zero.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-008, REQ-SYS-010, REQ-POINTER-001

Code Traces

- `include/juno/memory/pointer_api.h:188` (JunoMemory_PointerVerify)

Test Traces

- `tests/test_pointer.c:135` (test_pointer_verify_valid)
- `tests/test_pointer.c:143` (test_pointer_verify_null_api)
- `tests/test_pointer.c:151` (test_pointer_verify_null_addr)
- `tests/test_pointer.c:159` (test_pointer_verify_zero_size)

3.13.5. REQ-POINTER-005: Pointer Type Verification

Description

The pointer module shall verify that a pointer descriptor's size and alignment match an expected type.

Rationale

Enables deterministic, pool-based memory management for resource-constrained embedded systems.

Verification Method

Test

Uses (parent)

REQ-SYS-008, REQ-POINTER-004

Code Traces

- `include/juno/memory/pointer_api.h:129` (JunoMemory_PointerVerify)

Test Traces

- `tests/test_pointer.c:172` (test_pointer_verify_type_match)
- `tests/test_pointer.c:180` (test_pointer_verify_type_wrong_size)
- `tests/test_pointer.c:189` (test_pointer_verify_type_wrong_alignment)

3.14. Module: QUEUE

3.14.1. REQ-QUEUE-001: Queue Initialization

Description

The queue module shall initialize from a caller-provided backing array.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-009,

REQ-SYS-010, REQ-ARRAY-001

Code Traces

- `src/juno_buff_queue.c:106` (JunoDs_QueueInit)
- `include/juno/ds/queue_api.h:116` (JunoDs_QueueVerify)

Test Traces

- `tests/test_queue.c:209` (test_queue_init_nominal)
- `tests/test_queue.c:220` (test_queue_init_null_queue)
- `tests/test_queue.c:230` (test_queue_init_null_array)
- `tests/test_queue.c:237` (test_queue_init_zero_capacity)
- `tests/test_queue.c:729` (test_queue_verify_valid_queue)
- `tests/test_queue.c:739` (test_queue_verify_null_queue)
- `tests/test_queue.c:746` (test_queue_verify_invalid_api)
- `tests/test_queue.c:759` (test_queue_api_verify_valid)
- `tests/test_queue.c:769` (test_queue_api_verify_null)
- `tests/test_queue.c:776` (test_queue_api_verify_missing_functions)

3.14.2. REQ-QUEUE-002: Queue Initial State

Description

The queue shall set start index and length to zero on initialization.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-002, REQ-SYS-007, REQ-SYS-008, REQ-SYS-011, REQ-QUEUE-001

Code Traces

- `src/juno_buff_queue.c:106` (JunoDs_QueueInit)

Test Traces

- `tests/test_queue.c:209` (test_queue_init_nominal)

3.14.3. REQ-QUEUE-003: Queue Enqueue

Description

The queue shall copy an element to the back of the circular buffer in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-QUEUE-001

Code Traces

- `src/juno_buff_queue.c:24` (JunoDs_QueuePush)

Test Traces

- `tests/test_queue.c:249` (test_queue_push_single_item)
- `tests/test_queue.c:266` (test_queue_push_multiple_items)
- `tests/test_queue.c:290` (test_queue_push_to_full_capacity)
- `tests/test_queue.c:331` (test_queue_push_null_queue)
- `tests/test_queue.c:341` (test_queue_push_invalid_pointer)
- `tests/test_queue.c:554` (test_queue_circular_wraparound)
- `tests/test_queue.c:605` (test_queue_wraparound_at_boundary)
- `tests/test_queue.c:642` (test_queue_stress_push_pop_cycles)
- `tests/test_queue.c:673` (test_queue_interleaved_operations)
- `tests/test_queue.c:803` (test_queue_single_element_capacity)
- `tests/test_queue.c:831` (test_queue_boundary_index_calculation)
- `tests/test_queue.c:860` (test_queue_data_integrity_after_operations)

3.14.4. REQ-QUEUE-004: Queue Full Rejection

Description

The queue shall return JUNO_STATUS_INVALID_SIZE_ERROR when enqueue is attempted on a full queue.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-QUEUE-003

Code Traces

- `src/juno_buff_queue.c:24` (JunoDs_QueuePush)

Test Traces

- `tests/test_queue.c:308` (test_queue_push_overflow)

3.14.5. REQ-QUEUE-005: Queue Dequeue

Description

The queue shall remove and copy out the front element in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-QUEUE-001

Code Traces

- `src/juno_buff_queue.c:50` (JunoDs_QueuePop)

Test Traces

- `tests/test_queue.c:356` (test_queue_pop_single_item)
- `tests/test_queue.c:379` (test_queue_pop_multiple_items_fifo_order)
- `tests/test_queue.c:421` (test_queue_pop_null_queue)
- `tests/test_queue.c:430` (test_queue_pop_invalid_pointer)
- `tests/test_queue.c:554` (test_queue_circular_wraparound)
- `tests/test_queue.c:605` (test_queue_wraparound_at_boundary)
- `tests/test_queue.c:642` (test_queue_stress_push_pop_cycles)
- `tests/test_queue.c:673` (test_queue_interleaved_operations)
- `tests/test_queue.c:803` (test_queue_single_element_capacity)
- `tests/test_queue.c:860` (test_queue_data_integrity_after_operations)

3.14.6. REQ-QUEUE-006: Queue Empty Dequeue Rejection

Description

The queue shall return JUNO_STATUS_ERR when dequeue is attempted on an empty queue.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-QUEUE-005

Code Traces

- `src/juno_buff_queue.c:50` (JunoDs_QueuePop)

Test Traces

- `tests/test_queue.c:409` (test_queue_pop_empty_queue)

3.14.7. REQ-QUEUE-007: Queue Peek

Description

The queue shall return a pointer descriptor to the front element without removing it in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-QUEUE-001

Code Traces

- `src/juno_buff_queue.c:78` (JunoDs_QueuePeek)

Test Traces

- `tests/test_queue.c:452` (test_queue_peek_single_item)
- `tests/test_queue.c:478` (test_queue_peek_multiple_times)
- `tests/test_queue.c:515` (test_queue_peek_null_queue)
- `tests/test_queue.c:522` (test_queue_peek_after_pop)

3.14.8. REQ-QUEUE-008: Queue Empty Peek Rejection

Description

The queue shall return JUNO_STATUS_INVALID_SIZE_ERROR when peek is attempted on an empty queue.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-QUEUE-007

Code Traces

- `src/juno_buff_queue.c:78` (JunoDs_QueuePeek)

Test Traces

- `tests/test_queue.c:505` (test_queue_peek_empty_queue)

3.15. Module: SB

3.15.1. REQ-SB-001: Broker Root Structure

Description

The broker root shall hold a pipe registry pointer, registry length, and registry capacity.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-009

Code Traces

- `include/juno/sb/broker_api.h:79` (JUNO_MODULE_ROOT)

3.15.2. REQ-SB-002: Broker API Interface

Description

The broker API vtable shall define Publish and RegisterSubscriber operations.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SB-001

Code Traces

- `include/juno/sb/broker_api.h:89` (<unknown>)

3.15.3. REQ-SB-003: Broker Initialization

Description

The broker shall initialize from a caller-provided pipe registry array and capacity.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-002, REQ-SYS-007, REQ-SYS-008, REQ-SB-001

Code Traces

- `src/juno_broker.c:35` (JunoSb_BrokerInit)

Test Traces

- `tests/test_sb.c:190` (test_sb_broker_init_success)
- `tests/test_sb.c:203` (test_sb_broker_init_null_broker)
- `tests/test_sb.c:212` (test_sb_broker_init_null_registry)

3.15.4. REQ-SB-004: Pipe Derivation

Description

A pipe shall derive from the queue module root with an additional message ID field.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-MODULE-003, REQ-QUEUE-001

Code Traces

- `include/juno/sb/broker_api.h:72` (JUNO_MODULE_DERIVE)

3.15.5. REQ-SB-005: Pipe Initialization

Description

A pipe shall initialize with a message ID and a backing queue array.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SB-004, REQ-SYS-002

Code Traces

- `include/juno/sb/broker_api.h:138` (JunoSb_PipeInit)

Test Traces

- `tests/test_sb.c:225` (test_sb_pipe_init_success)
- `tests/test_sb.c:234` (test_sb_pipe_init_null_pipe)

3.15.6. REQ-SB-006: Publish Fan-Out

Description

The broker shall enqueue a message into every registered pipe whose message ID matches the published message ID.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SB-002

Code Traces

- `src/juno_broker.c:50` (Publish)

Test Traces

- `tests/test_sb.c:280` (test_sb_publish_fan_out)

3.15.7. REQ-SB-007: Publish Non-Matching Skip

Description

The broker shall skip pipes whose message ID does not match the published message ID.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SB-006

Code Traces

- `src/juno_broker.c:50` (Publish)

Test Traces

- `tests/test_sb.c:309` (test_sb_publish_skip_non_matching)

3.15.8. REQ-SB-008: Register Subscriber

Description

The broker shall append a pipe pointer to the registry on registration.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SB-002

Code Traces

- `src/juno_broker.c:75` (RegisterSubscriber)

Test Traces

- `tests/test_sb.c:246` (test_sb_register_subscriber)

3.15.9. REQ-SB-009: Registry Full Rejection

Description

The broker shall return an error status when registration is attempted on a full registry.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SB-008

Code Traces

- `src/juno_broker.c:75` (RegisterSubscriber)

Test Traces

- `tests/test_sb.c:259` (test_sb_register_full_rejection)

3.15.10. REQ-SB-010: Broker Vtable Validation

Description

The broker shall verify that its API vtable entries are non-null before each operation.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-010, REQ-SYS-011

Code Traces

- `include/juno/sb/broker_api.h:113` (JunoSb_BrokerVerify)

Test Traces

- `tests/test_sb.c:336` (test_sb_broker_verify_null_api)

3.15.11. REQ-SB-011: Broker Registry Validation

Description

The broker shall verify that its pipe registry pointer and capacity are valid before each operation.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-011

Code Traces

- `include/juno/sb/broker_api.h:113` (JunoSb_BrokerVerify)

Test Traces

- `tests/test_sb.c:346` (test_sb_broker_verify_null_registry)
- `tests/test_sb.c:356` (test_sb_broker_verify_zero_capacity)

3.15.12. REQ-SB-012: Pipe Validation

Description

The broker shall verify each pipe via the queue verification before publishing to it.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SB-006, REQ-SYS-008

Code Traces

- `include/juno/sb/broker_api.h:125` (JunoSb_PipeVerify)

Test Traces

- `tests/test_sb.c:366` (test_sb_pipe_verify_valid)
- `tests/test_sb.c:374` (test_sb_pipe_verify_null)

3.15.13. REQ-SB-013: Single Thread Scope

Description

The broker shall operate within a single thread without thread-safety mechanisms.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Code Traces

- `include/juno/sb/broker_api.h:50` (<unknown>)

3.16. Module: SCH

3.16.1. REQ-SCH-001: Scheduler Root Structure

Description

The scheduler root shall hold a time source pointer, a flattened schedule table, apps-per-minor-frame count, minor-frame count, and minor-frame period.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-006, REQ-SYS-007, REQ-SYS-009, REQ-SYS-010

Code Traces

- `include/juno/sch/juno_sch_api.h:61` (JUNO_MODULE_ROOT)

3.16.2. REQ-SCH-002: Scheduler API Interface

Description

The scheduler API vtable shall define Execute, GetMinorFramePeriod, and GetMajorFramePeriod operations.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SCH-001

Code Traces

- `include/juno/sch/juno_sch_api.h:70` (<unknown>)

3.16.3. REQ-SCH-003: Schedule Table Layout

Description

The schedule table shall be a flattened 2D array of application pointers with dimensions minor-frame count by apps-per-minor-frame.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SCH-001, REQ-SYS-002

Code Traces

- `include/juno/sch/juno_sch_api.h:56` (<unknown>)

3.16.4. REQ-SCH-004: Execute Major Frame

Description

The scheduler Execute operation shall invoke each application's OnProcess hook across all minor frames in one major frame.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SCH-002, REQ-APP-003

Code Traces

- `include/juno/sch/juno_sch_api.h:74` (JUNO_STATUS_T)

Test Traces

- `tests/test_sch.c:153` (test_sch_execute_invokes_all_apps)
- `tests/test_sch.c:162` (test_sch_execute_skips_null_slot)

3.16.5. REQ-SCH-005: Minor Frame Period Query

Description

The scheduler shall return the configured minor frame period as a timestamp.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SCH-002

Code Traces

- `include/juno/sch/juno_sch_api.h:77` (JUNO_TIMESTAMP_RESULT_T)

Test Traces

- `tests/test_sch.c:177` (test_sch_get_minor_frame_period)

3.16.6. REQ-SCH-006: Major Frame Period Computation

Description

The scheduler shall compute the major frame period as the product of the minor frame period and the minor-frame count.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SCH-005

Code Traces

- `include/juno/sch/juno_sch_api.h:80` (JUNO_TIMESTAMP_RESULT_T)

Test Traces

- `tests/test_sch.c:190` (test_sch_get_major_frame_period)

3.16.7. REQ-SCH-007: Time Source Dependency

Description

The scheduler shall use an injected time module for timing operations.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-007, REQ-TIME-002

Code Traces

- `include/juno/sch/juno_sch_api.h:61` (JUNO_MODULE_ROOT)

3.17. Module: SM

3.17.1. REQ-SM-001: SM Root Structure

Description

The state machine root shall hold a pointer to the current state.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-009, REQ-SYS-010

Code Traces

- `include/juno/sm/sm_api.h:76` (JUNO_MODULE_ROOT)

3.17.2. REQ-SM-002: State Root Structure

Description

Each state shall embed a module root with an API vtable, a pointer to the owning SM, and a next-state option.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SYS-006, REQ-MODULE-003

Code Traces

- `include/juno/sm/sm_api.h:59` (JUNO_MODULE_ROOT)

3.17.3. REQ-SM-003: State API Interface

Description

Each state API vtable shall define StateAction, ShouldExit, and ResetState operations.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Inspection

Uses (parent)

REQ-SM-002

Code Traces

- `include/juno/sm/sm_api.h:65` (JUNO_STATUS_T)

3.17.4. REQ-SM-004: SM Initialization

Description

The state machine shall initialize with a start state and failure handler.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-002, REQ-SYS-007, REQ-SM-001

Code Traces

- `include/juno/sm/sm_api.h:126` (JunoSm_Init)

Test Traces

- `tests/test_sm.c:91` (test_sm_init_success)
- `tests/test_sm.c:103` (test_sm_init_null_root)
- `tests/test_sm.c:110` (test_sm_init_null_start_state)

3.17.5. REQ-SM-005: State Initialization

Description

A state shall initialize with a link to its owning SM, an API vtable, and an optional next state.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SM-002, REQ-SYS-007

Code Traces

- `include/juno/sm/sm_api.h:104` (JunoSm_StateInit)

Test Traces

- `tests/test_sm.c:121` (test_sm_state_init_with_next)
- `tests/test_sm.c:136` (test_sm_state_init_without_next)
- `tests/test_sm.c:148` (test_sm_state_init_null_state)
- `tests/test_sm.c:156` (test_sm_state_init_null_api)

3.17.6. REQ-SM-006: Get Current State

Description

The state machine shall return the current state pointer.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SM-001

Code Traces

- `include/juno/sm/sm_api.h:139` (JunoSm_GetCurrentState)

Test Traces

- `tests/test_sm.c:168` (test_sm_get_current_state)
- `tests/test_sm.c:179` (test_sm_get_current_state_null_root)

3.17.7. REQ-SM-007: State Transition

Description

The state machine shall transition to the next state when the current state's next-state option is present.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SM-001

Code Traces

- `include/juno/sm/sm_api.h:151` (JunoSm_TransitionState)

Test Traces

- `tests/test_sm.c:190` (test_sm_transition_to_next)
- `tests/test_sm.c:220` (test_sm_transition_null_root)

3.17.8. REQ-SM-008: No Transition When None

Description

The state machine shall remain in the current state when the next-state option is absent.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SM-007

Code Traces

- `include/juno/sm/sm_api.h:151` (JunoSm_TransitionState)

Test Traces

- `tests/test_sm.c:207` (test_sm_transition_stays_when_none)

3.17.9. REQ-SM-009: SM Verification

Description

The state machine shall verify that its root and current state pointer are non-NULL.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Code Traces

- `include/juno/sm/sm_api.h:83` (JunoSm_Verify)

Test Traces

- `tests/test_sm.c:231` (test_sm_verify_valid)
- `tests/test_sm.c:241` (test_sm_verify_null_root)
- `tests/test_sm.c:248` (test_sm_verify_null_current_state)

3.17.10. REQ-SM-010: State Verification

Description

Each state shall verify that its API vtable and owning SM pointer are non-NULL.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-010, REQ-SYS-011

Code Traces

- `include/juno/sm/sm_api.h:91` (JunoSm_StateVerify)

Test Traces

- `tests/test_sm.c:260` (test_sm_state_verify_valid)
- `tests/test_sm.c:270` (test_sm_state_verify_null_state)
- `tests/test_sm.c:277` (test_sm_state_verify_null_api)
- `tests/test_sm.c:286` (test_sm_state_verify_null_sm)

3.17.11. REQ-SM-011: Next State Pointer Verification

Description

Each state shall verify that the next-state pointer is non-NULL when the next-state option is marked present.

Rationale

Provides a reusable application framework pattern for structured embedded software.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Code Traces

- `include/juno/sm/sm_api.h:91` (JunoSm_StateVerify)

Test Traces

- `tests/test_sm.c:295` (test_sm_state_verify_next_null_when_some)
- `tests/test_sm.c:306` (test_sm_state_verify_next_valid_when_some)

3.18. Module: STACK

3.18.1. REQ-STACK-001: Stack Initialization

Description

The stack module shall initialize from a caller-provided backing array.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-009, REQ-SYS-010, REQ-ARRAY-001

Code Traces

- `src/juno_buff_stack.c:101` (JunoDs_StackInit)

Test Traces

- `tests/test_stack.c:211` (test_stack_init_nominal)
- `tests/test_stack.c:221` (test_stack_init_null_stack)
- `tests/test_stack.c:228` (test_stack_init_null_api)
- `tests/test_stack.c:236` (test_stack_init_zero_capacity)
- `tests/test_stack.c:705` (test_stack_verify_valid_stack)
- `tests/test_stack.c:715` (test_stack_verify_null_stack)
- `tests/test_stack.c:722` (test_stack_verify_invalid_api)
- `tests/test_stack.c:735` (test_stack_verify_missing_push_function)
- `tests/test_stack.c:751` (test_stack_verify_missing_pop_function)

- `tests/test_stack.c:767` (test_stack_verify_missing_peek_function)
- `tests/test_stack.c:1170` (test_stack_init_with_array_verify_failure)

3.18.2. REQ-STACK-002: Stack Initial State

Description

The stack shall set length to zero on initialization.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-SYS-002, REQ-SYS-007, REQ-SYS-008, REQ-SYS-011, REQ-STACK-001

Code Traces

- `src/juno_buff_stack.c:101` (JunoDs_StackInit)

Test Traces

- `tests/test_stack.c:211` (test_stack_init_nominal)

3.18.3. REQ-STACK-003: Stack Push

Description

The stack shall copy an element to the top in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-STACK-001

Code Traces

- `src/juno_buff_stack.c:27` (JunoDs_StackPush)

Test Traces

- `tests/test_stack.c:248` (test_stack_push_single_item)
- `tests/test_stack.c:265` (test_stack_push_multiple_items)
- `tests/test_stack.c:289` (test_stack_push_to_full_capacity)
- `tests/test_stack.c:330` (test_stack_push_null_stack)

- `tests/test_stack.c:340` (test_stack_push_invalid_pointer)
- `tests/test_stack.c:578` (test_stack_stress_push_pop_cycles)
- `tests/test_stack.c:609` (test_stack_interleaved_operations)
- `tests/test_stack.c:667` (test_stack_fill_empty_cycles)
- `tests/test_stack.c:787` (test_stack_single_element_capacity)
- `tests/test_stack.c:820` (test_stack_data_integrity_after_operations)
- `tests/test_stack.c:848` (test_stack_alternating_push_pop)
- `tests/test_stack.c:966` (test_stack_push_at_exact_capacity)
- `tests/test_stack.c:1079` (test_stack_push_with_setAt_failure)

3.18.4. REQ-STACK-004: Stack Full Rejection

Description

The stack shall return JUNO_STATUS_INVALID_SIZE_ERROR when push is attempted on a full stack.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-STACK-003

Code Traces

- `src/juno_buff_stack.c:27` (JunoDs_StackPush)

Test Traces

- `tests/test_stack.c:307` (test_stack_push_overflow)
- `tests/test_stack.c:966` (test_stack_push_at_exact_capacity)

3.18.5. REQ-STACK-005: Stack Pop

Description

The stack shall remove and copy out the top element in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-STACK-001

Code Traces

- `src/juno_buff_stack.c:52` (JunoDs_StackPop)

Test Traces

- `tests/test_stack.c:355` (test_stack_pop_single_item)
- `tests/test_stack.c:378` (test_stack_pop_multiple_items_lifo_order)
- `tests/test_stack.c:420` (test_stack_pop_null_stack)
- `tests/test_stack.c:429` (test_stack_pop_invalid_pointer)
- `tests/test_stack.c:578` (test_stack_stress_push_pop_cycles)
- `tests/test_stack.c:609` (test_stack_interleaved_operations)
- `tests/test_stack.c:667` (test_stack_fill_empty_cycles)
- `tests/test_stack.c:787` (test_stack_single_element_capacity)
- `tests/test_stack.c:820` (test_stack_data_integrity_after_operations)
- `tests/test_stack.c:848` (test_stack_alternating_push_pop)
- `tests/test_stack.c:992` (test_stack_pop_until_empty)
- `tests/test_stack.c:1099` (test_stack_pop_with_getAt_failure)
- `tests/test_stack.c:1148` (test_stack_pop_copy_failure)

3.18.6. REQ-STACK-006: Stack Empty Pop Rejection

Description

The stack shall return JUNO_STATUS_INVALID_SIZE_ERROR when pop is attempted on an empty stack.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-STACK-005

Code Traces

- `src/juno_buff_stack.c:52` (JunoDs_StackPop)

Test Traces

- `tests/test_stack.c:408` (test_stack_pop_empty_stack)
- `tests/test_stack.c:992` (test_stack_pop_until_empty)

3.18.7. REQ-STACK-007: Stack Peek

Description

The stack shall return a pointer descriptor to the top element without removing it in O(1) time.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-STACK-001

Code Traces

- `src/juno_buff_stack.c:77` (JunoDs_StackPeek)

Test Traces

- `tests/test_stack.c:451` (test_stack_peek_single_item)
- `tests/test_stack.c:477` (test_stack_peek_top_element)
- `tests/test_stack.c:501` (test_stack_peek_multiple_times)
- `tests/test_stack.c:538` (test_stack_peek_null_stack)
- `tests/test_stack.c:545` (test_stack_peek_after_pop)
- `tests/test_stack.c:880` (test_stack_peek_consistency_during_operations)
- `tests/test_stack.c:1125` (test_stack_peek_with_getAt_failure)

3.18.8. REQ-STACK-008: Stack Empty Peek Rejection

Description

The stack shall return JUNO_STATUS_INVALID_SIZE_ERROR when peek is attempted on an empty stack.

Rationale

Enables fundamental data structure capabilities for embedded applications without dynamic allocation.

Verification Method

Test

Uses (parent)

REQ-STACK-007

Code Traces

- `src/juno_buff_stack.c:77` (JunoDs_StackPeek)

Test Traces

- `tests/test_stack.c:528` (test_stack_peek_empty_stack)

3.19. Module: STATUS

3.19.1. REQ-STATUS-001: Unified Status Type

Description

The status module shall define a single 32-bit integer status type (JUNO_STATUS_T) for error propagation.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-003, REQ-SYS-004, REQ-SYS-009

Code Traces

- `include/juno/status.h:50` (<unknown>)

3.19.2. REQ-STATUS-002: Canonical Status Codes

Description

The status module shall define distinct status codes for each error condition category (success, null pointer, memory, type, size, table full, does-not-exist, I/O, CRC, reference, data, timeout, out-of-bounds).

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-004

Code Traces

- `include/juno/status.h:57` (<unknown>)

3.19.3. REQ-STATUS-003: Failure Handler Callback Type

Description

The status module shall define an optional failure handler callback type that receives a status code, message, and user data.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-SYS-004

Code Traces

- `include/juno/status.h:107` (<unknown>)

3.19.4. REQ-STATUS-004: Failure Handler Non-Interference

Description

The failure handler shall not alter control flow when invoked.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Uses (parent)

REQ-STATUS-003

Code Traces

- `include/juno/status.h:127` (<unknown>)

3.20. Module: SYS

3.20.1. REQ-SYS-001: Zero Dynamic Allocation

Description

The library shall not use malloc, calloc, realloc, or free.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-APP-001, REQ-ARRAY-001, REQ-CRC-001, REQ-HASH-001, REQ-HEAP-001, REQ-IO-001, REQ-LOG-001, REQ-MAP-001, REQ-MATH-001, REQ-MEMORY-001, REQ-MODULE-001, REQ-MTX-001, REQ-POINTER-001, REQ-QUEUE-001, REQ-SB-001, REQ-SCH-001, REQ-SM-001, REQ-STACK-001, REQ-STATUS-001, REQ-TIME-001

Code Traces

- `CMakeLists.txt:75` (list)

3.20.2. REQ-SYS-002: Caller-Owned Memory

Description

All memory used by the library shall be caller-owned and injected via initialization functions.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-APP-001, REQ-ARRAY-001, REQ-HEAP-001, REQ-IO-001, REQ-LOG-006, REQ-MAP-001, REQ-MEMORY-003, REQ-MTX-001, REQ-QUEUE-001, REQ-QUEUE-002, REQ-SB-003, REQ-SB-005, REQ-SCH-001, REQ-SCH-003, REQ-SM-004, REQ-STACK-001, REQ-STACK-002, REQ-TIME-001

Code Traces

- `include/juno/module.h:54` (JUNO_MODULE_DECLARE)

3.20.3. REQ-SYS-003: Freestanding C11 Compatibility

Description

The library shall compile under -std=c11 -nostdlib -ffreestanding without hosted C standard library dependencies.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Analysis

Implements (children)

REQ-APP-001, REQ-ARRAY-001, REQ-CRC-001, REQ-HASH-001, REQ-HEAP-001, REQ-IO-001, REQ-LOG-001, REQ-MAP-001, REQ-MATH-001, REQ-MATH-002, REQ-MATH-004, REQ-MEMORY-001,

REQ-MODULE-001, REQ-MTX-001, REQ-POINTER-001, REQ-QUEUE-001, REQ-SB-001, REQ-SCH-001, REQ-SM-001, REQ-STACK-001, REQ-STATUS-001, REQ-TIME-001

Code Traces

- `CMakeLists.txt:75` (list)

3.20.4. REQ-SYS-004: Explicit Status Code Returns

Description

All fallible functions shall return explicit status codes (JUNO_STATUS_T).

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-HASH-001, REQ-HASH-003, REQ-HEAP-001, REQ-MAP-001, REQ-MEMORY-001, REQ-MODULE-005, REQ-MODULE-006, REQ-POINTER-001, REQ-QUEUE-001, REQ-SB-001, REQ-SM-001, REQ-STACK-001, REQ-STATUS-001, REQ-STATUS-002, REQ-STATUS-003, REQ-TIME-001

Code Traces

- `include/juno/status.h:40` (<unknown>)

3.20.5. REQ-SYS-005: No Silent Failures

Description

The library shall not silently swallow errors or produce undefined behavior on error paths.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-CRC-002, REQ-CRC-003, REQ-HASH-001, REQ-HEAP-001, REQ-MAP-001, REQ-MEMORY-001, REQ-POINTER-001, REQ-QUEUE-001, REQ-SB-001, REQ-SM-001, REQ-STACK-001, REQ-TIME-001, REQ-TIME-018

Code Traces

- `include/juno/status.h:40` (<unknown>)

3.20.6. REQ-SYS-006: Dependency Injection via Vtables

Description

All modules shall use the vtable/module-root/derivation pattern for dependency injection.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-APP-001, REQ-ARRAY-001, REQ-HEAP-001, REQ-IO-001, REQ-IO-009, REQ-IO-013, REQ-LOG-001, REQ-MAP-001, REQ-MEMORY-001, REQ-MODULE-001, REQ-MODULE-003, REQ-MODULE-004, REQ-MTX-001, REQ-POINTER-001, REQ-QUEUE-001, REQ-SB-001, REQ-SCH-001, REQ-SM-001, REQ-SM-002, REQ-STACK-001, REQ-TIME-001, REQ-TIME-002

Code Traces

- `include/juno/module.h:54` (JUNO_MODULE_DECLARE)

3.20.7. REQ-SYS-007: No Global Dependencies

Description

Dependencies shall be injected via initialization functions, never globally referenced.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-APP-001, REQ-ARRAY-001, REQ-HEAP-001, REQ-HEAP-002, REQ-IO-001, REQ-LOG-006, REQ-MAP-001, REQ-MAP-002, REQ-MAP-003, REQ-MEMORY-002, REQ-MEMORY-003, REQ-MTX-001, REQ-QUEUE-002, REQ-SB-003, REQ-SCH-001, REQ-SCH-007, REQ-SM-004, REQ-SM-005, REQ-STACK-002, REQ-TIME-001

Code Traces

- `include/juno/macros.h:50` (<unknown>)

3.20.8. REQ-SYS-008: Precondition Verification

Description

All public API functions shall verify their preconditions at entry.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Test

Implements (children)

REQ-ARRAY-005, REQ-ARRAY-007, REQ-HASH-001, REQ-HASH-002, REQ-HEAP-001, REQ-MAP-001, REQ-MEMORY-003, REQ-POINTER-004, REQ-POINTER-005, REQ-QUEUE-002, REQ-SB-003, REQ-SB-012, REQ-SM-009, REQ-SM-011, REQ-STACK-002, REQ-TIME-001, REQ-TIME-019

Code Traces

- `include/juno/macros.h:50` (<unknown>)
- `include/juno/macros.h:88` (<unknown>)

Test Traces

- `tests/test_pointer.c:151` (test_pointer_verify_null_addr)

3.20.9. REQ-SYS-009: No Global Mutable State

Description

The library shall contain no global mutable state.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Inspection

Implements (children)

REQ-APP-001, REQ-ARRAY-001, REQ-CRC-001, REQ-HASH-001, REQ-HEAP-001, REQ-IO-001, REQ-LOG-001, REQ-MAP-001, REQ-MATH-001, REQ-MEMORY-001, REQ-MODULE-001, REQ-MTX-001, REQ-POINTER-001, REQ-QUEUE-001, REQ-SB-001, REQ-SCH-001, REQ-SM-001, REQ-STACK-001, REQ-STATUS-001, REQ-TIME-001

Code Traces

- `include/juno/module.h:54` (JUNO_MODULE_DECLARE)

3.20.10. REQ-SYS-010: API Vtable Validation

Description

All modules shall verify that their API vtable and all function pointers within it reference valid (non-NULL) memory before performing any operation.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without

hosted OS assumptions.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Implements (children)

REQ-APP-001, REQ-ARRAY-006, REQ-HEAP-007, REQ-IO-001, REQ-LOG-001, REQ-MAP-009, REQ-MEMORY-001, REQ-MTX-005, REQ-POINTER-004, REQ-QUEUE-001, REQ-SB-010, REQ-SCH-001, REQ-SM-001, REQ-SM-010, REQ-STACK-001, REQ-TIME-001

Code Traces

- `include/juno/macros.h:50` (<unknown>)

Test Traces

- `tests/test_array.c:238` (test_array_verify_null_api)

3.20.11. REQ-SYS-011: Injected Dependency Validation

Description

All modules shall verify that all injected dependency pointers are valid (non-NULL) before performing any operation.

Rationale

Required to ensure LibJuno operates reliably on bare-metal and freestanding targets without hosted OS assumptions.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Implements (children)

REQ-HEAP-008, REQ-MAP-010, REQ-MEMORY-003, REQ-QUEUE-002, REQ-SB-010, REQ-SB-011, REQ-SM-010, REQ-STACK-002, REQ-TIME-001

Code Traces

- `include/juno/macros.h:50` (<unknown>)

Test Traces

- `tests/test_sb.c:346` (test_sb_broker_verify_null_registry)

3.21. Module: TIME

3.21.1. REQ-TIME-001: Timestamp Representation

Description

The time module shall represent time as a struct with integer seconds and fixed-point fractional subseconds fields.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Inspection

Uses (parent)

REQ-SYS-001, REQ-SYS-002, REQ-SYS-003, REQ-SYS-004, REQ-SYS-005, REQ-SYS-006, REQ-SYS-007, REQ-SYS-008, REQ-SYS-009, REQ-SYS-010, REQ-SYS-011

Code Traces

- `include/juno/time/time_api.h:104` (<unknown>)

3.21.2. REQ-TIME-002: Time API Interface

Description

The time module shall define a vtable-based API with Now, AddTime, SubtractTime, SleepTo, Sleep, and conversion operations.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Inspection

Uses (parent)

REQ-SYS-006

Code Traces

- `include/juno/time/time_api.h:128` (JUNO_TIMESTAMP_RESULT_T)

3.21.3. REQ-TIME-003: AddTime Operation

Description

The time module shall add a duration to a timestamp in-place.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:26` (JunoTime_AddTime)

Test Traces

- `tests/test_time.c:86` (test_AddTime_no_subseconds_overflow)

3.21.4. REQ-TIME-004: AddTime Subsecond Carry-Over

Description

The time module shall carry over subseconds to seconds when subsecond addition overflows.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-003

Code Traces

- `src/juno_time.c:26` (JunoTime_AddTime)

Test Traces

- `tests/test_time.c:97` (test_AddTime_with_subseconds_overflow)

3.21.5. REQ-TIME-005: SubtractTime Operation

Description

The time module shall subtract a duration from a timestamp in-place.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:49` (JunoTime_SubtractTime)

Test Traces

- `tests/test_time.c:110` (test_SubtractTime_no_borrow)

3.21.6. REQ-TIME-006: SubtractTime Subsecond Borrow

Description

The time module shall borrow from seconds when subsecond subtraction underflows.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-005

Code Traces

- `src/juno_time.c:49` (JunoTime_SubtractTime)

Test Traces

- `tests/test_time.c:122` (test_SubtractTime_borrow_subseconds)

3.21.7. REQ-TIME-007: SubtractTime Negative Saturation

Description

The time module shall saturate the result to zero seconds and zero subseconds when subtraction would produce negative time.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-005

Code Traces

- `src/juno_time.c:49` (JunoTime_SubtractTime)

Test Traces

- `tests/test_time.c:135` (test_SubtractTime_invalid_lower_seconds)
- `tests/test_time.c:146` (test_SubtractTime_invalid_zero_seconds_insufficient_subseconds)

3.21.8. REQ-TIME-008: SubtractTime Negative Error

Description

The time module shall return JUNO_STATUS_INVALID_DATA_ERROR when subtraction would produce negative time.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-007

Code Traces

- `src/juno_time.c:49` (JunoTime_SubtractTime)

Test Traces

- `tests/test_time.c:135` (test_SubtractTime_invalid_lower_seconds)
- `tests/test_time.c:146` (test_SubtractTime_invalid_zero_seconds_insufficient_subseconds)

3.21.9. REQ-TIME-009: Timestamp to Nanoseconds Conversion

Description

The time module shall convert a timestamp to nanoseconds using half-up rounding.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:78` (JunoTime_TimestampToNanos)

Test Traces

- `tests/test_time.c:157` (test_TimestampToNanos_success_integer)
- `tests/test_time.c:166` (test_TimestampToNanos_success_fractional)

3.21.10. REQ-TIME-010: Timestamp to Microseconds Conversion

Description

The time module shall convert a timestamp to microseconds using half-up rounding.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:115` (JunoTime_TimestampToMicros)

Test Traces

- `tests/test_time.c:177` (test_TimestampToMicros_success_integer)
- `tests/test_time.c:186` (test_TimestampToMicros_success_fractional)

3.21.11. REQ-TIME-011: Timestamp to Milliseconds Conversion

Description

The time module shall convert a timestamp to milliseconds using half-up rounding.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:149` (JunoTime_TimestampToMillis)

Test Traces

- `tests/test_time.c:196` (test_TimestampToMillis_success_integer)
- `tests/test_time.c:205` (test_TimestampToMillis_success_fractional)

3.21.12. REQ-TIME-012: Nanoseconds to Timestamp Conversion

Description

The time module shall convert nanoseconds to a timestamp with integer seconds and fractional subseconds.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:184` (JunoTime_NanosToTimestamp)

Test Traces

- `tests/test_time.c:215` (test_NanosToTimestamp_zero)
- `tests/test_time.c:225` (test_NanosToTimestamp_integer_and_fractional)

3.21.13. REQ-TIME-013: Microseconds to Timestamp Conversion

Description

The time module shall convert microseconds to a timestamp with integer seconds and fractional subseconds.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:202` (JunoTime_MicrosToTimestamp)

Test Traces

- `tests/test_time.c:240` (test_MicrosToTimestamp_zero)
- `tests/test_time.c:250` (test_MicrosToTimestamp_integer_and_fractional)

3.21.14. REQ-TIME-014: Milliseconds to Timestamp Conversion

Description

The time module shall convert milliseconds to a timestamp with integer seconds and fractional subseconds.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:220` (JunoTime_MillisToTimestamp)

Test Traces

- `tests/test_time.c:265` (test_MillisToTimestamp_zero)
- `tests/test_time.c:275` (test_MillisToTimestamp_integer_and_fractional)

3.21.15. REQ-TIME-015: Timestamp to Double Conversion

Description

The time module shall convert a timestamp to a double-precision floating-point seconds value.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:239` (JunoTime_TimestampToDouble)

Test Traces

- `tests/test_time.c:314` (test_timestamp_to_double)

3.21.16. REQ-TIME-016: Double to Timestamp Conversion

Description

The time module shall convert a non-negative double-precision seconds value to a timestamp.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-002

Code Traces

- `src/juno_time.c:254` (JunoTime_DoubleToTimestamp)

Test Traces

- `tests/test_time.c:324` (test_double_to_timestamp_success)
- `tests/test_time.c:333` (test_double_to_timestamp_zero)

3.21.17. REQ-TIME-017: Double to Timestamp Range Rejection

Description

The time module shall return JUNO_STATUS_INVALID_DATA_ERROR for negative or out-of-range double inputs.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-TIME-016

Code Traces

- `src/juno_time.c:254` (JunoTime_DoubleToTimestamp)

Test Traces

- `tests/test_time.c:342` (test_double_to_timestamp_negative_rejected)

3.21.18. REQ-TIME-018: Conversion Overflow Detection

Description

The time module shall return JUNO_STATUS_INVALID_DATA_ERROR when a timestamp-to-integer conversion would overflow.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-SYS-005

Code Traces

- `src/juno_time.c:78` (JunoTime_TimestampToNanos)
- `src/juno_time.c:115` (JunoTime_TimestampToMicros)
- `src/juno_time.c:149` (JunoTime_TimestampToMillis)

Test Traces

- `tests/test_time.c:349` (test_timestamp_to_nanos_overflow)

3.21.19. REQ-TIME-019: Null Pointer Rejection

Description

The time module shall return JUNO_STATUS_NULLPTR_ERROR when the time root pointer is NULL.

Rationale

Enables deterministic time arithmetic and conversion for real-time embedded applications.

Verification Method

Test

Uses (parent)

REQ-SYS-008

Code Traces

- `src/juno_time.c:78` (JunoTime_TimestampToNanos)
- `src/juno_time.c:115` (JunoTime_TimestampToMicros)
- `src/juno_time.c:149` (JunoTime_TimestampToMillis)
- `src/juno_time.c:184` (JunoTime_NanosToTimestamp)
- `src/juno_time.c:202` (JunoTime_MicrosToTimestamp)
- `src/juno_time.c:220` (JunoTime_MillisToTimestamp)
- `src/juno_time.c:239` (JunoTime_TimestampToDouble)
- `src/juno_time.c:254` (JunoTime_DoubleToTimestamp)

Test Traces

- `tests/test_time.c:362` (test_double_to_timestamp_null_root)

4. Traceability Validation Summary

 All traceability checks passed.